Machines for Babbage's Hall

Every machine considered, ranked by interest · 31 October 2026

Every non-electronic computing machine worth considering for the hall, from the list we began with and its additions (the conversation as exported, in *Mechanical computation machines*), the three papers (*Early Computing Machines*, *Computing after Babbage*, *Ball-and-disk integrator*) and the rest of the field. Clocks are left out (they are a collection of their own), and so are machines that compute with relays, valves or flowing water (see the end).

Each is scored from 1 to 5 for **interest** (how much there is to see and understand: surprise, motion, history) and for **complexity** (how hard to simulate in bpp and to print and build). The list ranks them by interest as I judge it, the most interesting first; the scores group them, and the order within a score is the judgement. The hall's bays follow the same order, walked the other way: the least interesting at the entrance, the Difference Engine at the far end.

Changes since the first list: Lehmer's sieve was scored 5 there (with a ball that drops when the holes line up), then 3; it is 4 here. The Curta and the Enigma were left out there (too dense; electromechanical) and are in here: the Curta as a cutaway, the Enigma for its mechanical stepping.

Status: *built*: in the hall, with its STL files (*built (accumulator)*: its first part); *bay*: has a covered bay "in preparation"; blank: not yet planned.

The ranked list

#	Machine	Year	What it does	In bpp, and printed	Int.	Cx.	Status
1	Babbage's Difference Engine	1822 / 1849	tabulates a polynomial by adding its differences, with no multiplying	two orders of difference, three digits, is a good model: figure wheels and Babbage's carry, after the Pascaline	5	5	bay
2	Antikythera mechanism	c. 100 BCE	a gear train for the Sun, Moon and planets, the lunar phase, eclipse dials; a pin-and-slot gear for the Moon's varying speed	gears, as in the Differential (convex-hull teeth), and many of them; printed replicas exist	5	4	
3	Enigma (its rotors' stepping)	1923	a keypress steps the rotors by pawls and notches; the current's path through them lights the cipher letter	the stepping is mechanical (the middle rotor's double step comes of itself); the wiring is a lookup in Lua; prints, with wire, contacts and LEDs (first left out as electromechanical)	5	3	
4	Differential Analyser	1931	wheel-and-disc integrators, wired in a loop, solve differential equations and draw the answer	rolling contact that must not slip; Bush's torque amplifiers can be motors; Hartree built one in Meccano	5	4	bay

#	Machine	Year	What it does	In bpp, and printed	Int.	Cx.	Status
5	Zuse Z1	1938	the first freely programmable computer, binary floating point, all mechanical: sliding steel plates and pins	its plate logic is lovely to simulate gate by gate; the whole machine is far beyond us	5	5	
6	Norden bombsight	1930s	a gyro-steadied sight whose analogue computer (rate discs, cams) solves the bombing problem	the computing (disc integrators and cams) is doable; the gyro too, as in Euler's Gallery; whole, very complex	5	5	
7	Ford Rangekeeper Mk I	1917	a battleship's fire-control computer: integrators, differentials and cams solve where the target will be	its parts each are doable (the Differential, integrators, cams); together, a big job	5	5	
8	Curta	1948	a pocket four-function calculator: one stepped drum, nines' complements for subtraction	small and dense: the stepped drum and carry are doable, the whole machine very hard; best as an oversized cutaway (first left out as too dense)	5	4	
9	Kelvin's tide predictor	1872	adds ten or more sine waves (the Sun's and Moon's tides) and draws a year of tides	cranks and slotted yokes are easy; the cord over the pulleys that adds them: bpp has no rope, but a rope of linked beads works if it isn't much lighter than its weights (tested: see below)	5	3	bay
10	Digi-Comp II	1965	marbles rolling down a board of flip-flops count, add and multiply in binary	built: a two-level board, thirteen printed rockers tipped over by the balls alone (a steel ball in each for weight, a foot below for CLEAR) make the accumulator, MQ and distributor; hand-set switches (MULTIPLY, CLEAR, COUNT, the memory), COMPLEMENT's shutters, OVERFLOW's; a feeder held by an escapement; AUTO, a thread from a trip lever at the bottom to the START lever; programs run line by line	5	3	built (count, add, multiply, subtract, divide; AUTO)
11	Domino Half Adder	modern	dominoes falling add two bits	built: 268 dominoes on printed tiles	4	2	built
12	Pascaline	1642	adds, with Pascal's falling-weight carry	built: three digits, sautoirs, detents, stylus and stop	4	3	built
13	Galton board	1873	balls falling through pins pile up into the normal curve	balls and pins only; easy, and a good opener	4	1	
14	Planimeter (Amsler's polar)	1856	trace a shape's outline and a rolling wheel reads its area	one wheel rolling and slipping on paper; small and clear	4	2	
15	Ball-and-disk integrator	1876	James Thomson's: a ball between a turning disc and a cylinder; the cylinder turns as the integral	the ball must roll without slipping; the integrator the tide predictor's brother and the analyser's ancestor	4	2	
16	Integraph	1880s	follows a curve and draws its integral	a wheel steering by its knife edge; a small step on from the planimeter	4	3	

#	Machine	Year	What it does	In bpp, and printed	Int.	Cx.	Status
17	"Basic Mechanisms" kit	1953	the US Navy fire-control course's set: differential, cam function generator, slide multiplier, component solver	each a small exhibit; together, one bay that explains the aiming machines	4	2	
18	Kelvin's harmonic analyser	1878	integrators find the Fourier coefficients of a measured curve (the tide predictor in reverse)	ball-and-disk integrators in a row	4	3	
19	Thomas Arithmometer	1820	the first mass-produced calculator: stepped drums, a sliding carriage	a stepped drum per digit and a carry; a fine follow-on to the Pascaline	4	3	
20	Leibniz's stepped reckoner	1694	the first four-function calculator: the stepped drum	the stepped drum is the heart of the next three machines	4	3	
21	Odhner pinwheel calculator (Brunsviga, Facit)	1873	pinwheels with retractable pins set the figures; turn the crank	flat pinwheels print well; the carry is the work	4	3	
22	Automatic Totalisator	1913	racetrack betting: a great shaft adder of differentials counts tickets and sets the odds	differentials (built) chained; the first real-time computer, the Science Museum says	4	3	
23	Hagelin M-209	1940	a fully mechanical cipher machine: six pin wheels and a drum of 27 lugged bars	all mechanical, but many small parts: pins, lugs, a print wheel	4	5	
24	Torres Quevedo's algebraic machines	1890s	find the roots of polynomials with his "endless spindle" logarithmic gearing	special gearing; little known and striking	4	4	
25	Kelvin's equation solver (and Wilbur's)	1878 / 1936	tilting plates on pulleys and cords solve simultaneous linear equations	plates, pulleys, cords: the cords are the hard part	4	3	
26	Michelson and Stratton's harmonic analyser	1898	80 springs sum and analyse waves; it showed the Gibbs overshoot	springs and levers: Bullet springs, many of them	4	3	
27	Comptometer	1887	key-driven adder: press a key and the column moves at once	keys, sectors and carry; no crank	4	3	
28	Digi-Comp I	1963	plastic sliders and pegs: three flip-flops and programmable logic	slides, pegs and a lever: simple parts, many of them	4	2	
29	Dr. Nim	1965	a marble machine that plays Nim, and wins	a few flip-flops and marbles: a little cousin of the Digi-Comp II	4	2	
30	Marble adders (Turing Tumble, Wandel's)	modern	marbles flip levers to add binary numbers	contact and gravity; very like the Digi-Comp II	4	2	

#	Machine	Year	What it does	In bpp, and printed	Int.	Cx.	Status
31	Argo clock and Dreyer table	1912 / 1911	WWI battleship fire control: plotting the target's course and range by integrators and a plotting table	integrators (as the ball-and-disk) and a plotting pen	4	4	
32	Torpedo Data Computer	1940s	a submarine's analogue fire control	like the rangekeeper, smaller; partly electrical	4	5	
33	Lehmer's number sieve	1926 / 1932	factors numbers: loops of different lengths (bicycle chains, later toothed wheels) turn until gaps line up, and a ball or pin drops through to stop it	built: four Geneva-driven wheels of 9, 8, 7 and 5 places with pegged holes, and a cam-lifted probe; factors by Fermat's way	4	3	built
34	The Differential	1827	gears that add and subtract (Pecqueur's patent; older in clocks)	built: printed bevel gears	3	2	built
35	Schickard's calculating clock	1623	the first known calculator: adding wheels and Napier's bones	a one-tooth carry gear (it jams on long carries: worth showing beside the Pascaline)	3	3	
36	Jacquard's card reader	1804	punched cards choose which hooks lift: the first stored program	needles, cards and hooks; a reader for a programmable exhibit	3	3	
37	Millionaire	1893	the first direct multiplier: a "multiplication body" of tongues gives each product at once	complex, dense	3	5	
38	Scheutz difference engine	1853	the first working difference engine, with a printer	as Babbage's; one of them is enough	3	5	
39	Three-dimensional cams	1930s	a cam shaped by two inputs gives a function of both (fire control's tables)	one shaped cam and a follower; prints well	3	2	
40	Petrol-pump price computer	1930s	multiplies the litres pumped by the price per litre, continuously, with variable-ratio gearing	a variator and counters; small	3	2	
41	Linkage function generators (the Peaucellier–Lipkin inverter and others)	1864 / 1900s	bar linkages that compute a product, a square, a log; Peaucellier's computes 1/x	a few hinged bars each; a set of them would make one bay	3	2	
42	Bead sort and spaghetti sort	modern	physical sorting: beads fall on rods, or rods are stood on end	beads and rods under gravity: easy	3	1	
43	Slide adder (troncet, Addiator)	1889 / 1920	slides moved with a stylus add and carry by a hook	flat slides; the carry hook is the trick	3	2	
44	Mechanical combination lock		the wheels' notches must line up: a lock that tests a number	wheels and a fence; simple	2	2	

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45	Napier's bones, Genaille–Lucas rulers	1617 / 1891	rods laid side by side give a product to read off	no moving mechanism: printable, nothing to simulate	2	1	
46	Slide rule	1620s	multiplies by adding logarithms	one slide; nothing to simulate	2	1	
47	Abacus, suanpan	ancient	beads	nothing to simulate	1	1	

By kind

- **Adding and counting** (digital): Schickard, the Pascaline, Leibniz, the Arithmometer, the pinwheels, the Comptometer, the Millionaire, the Curta, the slide adder, the Totalisator, the petrol-pump computer.
- **Tabulating and programmed:** the Difference Engines (Babbage's, Scheutz's), Jacquard's reader, the Zuse Z1.
- **Logic and play:** the Digi-Comps, Dr. Nim, marble adders, the domino adder, the Galton board, the bead and spaghetti sorts.
- **Number theory, ciphers and locks:** Lehmer's sieve, the Enigma, the Hagelin M-209, the combination lock.
- **Calculus and waves** (analogue): the planimeter, the integraph, the ball-and-disk integrator, Kelvin's tide predictor and harmonic analyser, Michelson and Stratton's analyser, the Differential Analyser.
- **Algebra:** Kelvin's and Wilbur's equation solvers, Torres Quevedo's root finders, linkage function generators, three-dimensional cams, the "Basic Mechanisms" kit.
- **Aiming:** the Argo clock and Dreyer table, the rangekeeper, the Norden bombsight, the Torpedo Data Computer.
- **Astronomy:** the Antikythera mechanism.

Left out, and why

- **Electromechanical and electronic:** Stibitz's Model K (relays), the bombes, the Harvard Mark I, Colossus, ENIAC and after, and the Mallock machine (electrical). The Enigma is kept because its stepping is mechanical and its wiring is a fixed lookup; the bombe is not, being relays and motors.
- **Water:** Lukyanov's water integrator and Phillips's MONIAC compute with flowing water, which Bullet does not do (rigid and soft bodies only).
- **Clocks and orreries:** a collection of their own.
- **Program stores that don't compute:** music boxes and Jacquard's looms (Jacquard's card reader is kept, as an input for another machine).
- **Hollerith's machines** (electrical counting) and **MEMS and nanomachines** (too small to print).

How the hall uses this list

The hall's bays are walked in rising interest, so a new exhibit goes in its place by its interest score. With eight bays, the hall holds the five built (Lehmer's sieve second, after the Differential; the Digi-Comp II fifth) and the three with bays; the Enigma would make a ninth (a fifth row of bays, one spare).

Rope. bpp has no rope (its soft bodies are cloth). A rope of small spheres joined by ball joints works, tested as an Atwood machine over a 5 cm pulley: with 60 beads making a 50 g rope between 200 g and 250 g weights, it stretched under 0.3% and the weights accelerated at 95 to 103 cm/s^2 against the

textbook 98, at 2400 steps a second and 100 solver passes, about a second of computing per simulated second. A rope 2500 times lighter per bead than its weights broke at any step size: the rope must not be too much lighter than what hangs on it. That suits the tide predictor (its pen weight can be light) and the equation solvers. The other way is to add Bullet's own rope (`btSoftBodyHelpers::CreateRope`) to bpp, a C++ change. The best candidates for new bays, for interest against effort, are the Galton board and the planimeter (quick and clear), and the Enigma and the Antikythera mechanism (striking, and within reach).